

VENKATA SAI KRISHNA ADITYA VATTURI

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PROFESSIONAL SUMMARY

Backend Engineer with 3 years of experience designing and optimizing REST APIs using Java (Spring Boot) and FastAPI. Improved API performance by up to 30% through caching and query optimization. Experienced in integrating LLM-based systems using RAG and vector search while maintaining production-grade reliability.

TECHNICAL SKILLS

Languages: Java, Python, SQL, JavaScript
Backend & APIs: Spring Boot, FastAPI, RESTful Microservices, MVC, WebSockets, Rate Limiting
AI & Data: RAG, LLM APIs (OpenAI/Groq), Vector Search, Semantic Caching
Databases: PostgreSQL, MongoDB, Redis, SQLite, Query Optimization, Indexing
Cloud & DevOps: Docker, CI/CD Pipelines, Git, Linux, Postman, Unit/Integration Testing

EXPERIENCE

Graduate Research Assistant Oct 2025 – Present
California State University, East Bay Hayward, CA

- **Designed and implemented reproducible ML lifecycles** (data prep to evaluation), deploying models as production ready FastAPI services for departmental research applications.
- **Built a comprehensive LLM evaluation harness** to benchmark prompt strategies and retrieval accuracy, directly increasing response reliability and documentation standards.
- **Enhanced system observability** by implementing structured logging and validation frameworks, reducing failure debugging time for research APIs.

AI Engineering Intern June 2025 – August 2025
Scholarship Auditions (Accoladi) California, USA

- **Engineered and deployed a production-ready RAG chatbot** utilizing Gemma-7B via Groq API, automating FAQ resolution and significantly reducing manual support load.
- **Reduced average query latency by 30%** by designing and implementing a vector-based semantic search layer and intelligent caching strategy.
- **Developed automated data ingestion pipelines** to ensure real-time synchronization between structured retrieval sources and LLM context windows.

Software Developer June 2022 – Dec 2024
Virtusa & YahvaAI Hyderabad, India

- **Optimized backend SQL queries and integrated Redis caching**, resulting in a **25% improvement in API response times** for enterprise-scale Java applications.
- **Designed and maintained secure REST APIs** using Spring Boot, implementing JWT-based authentication and fine-grained Role-Based Access Control (RBAC).
- **Spearheaded the refactoring of legacy modules** and increased test coverage through unit/integration testing, significantly reducing technical debt and improving maintainability.

PROJECTS

AI Interview Platform — *FastAPI, React, PostgreSQL*

- Developed an end-to-end evaluation system featuring speech-to-text scoring and automated rubric-based grading for candidate screening.
- Designed secure session management and admin dashboards for structured, multi-tenant candidate review.

Vite-Ledger — *FastAPI, React, PostgreSQL*

- Designed modular backend with FastAPI microservices and clear API contracts.
- Integrated LLM-based recommendation engine for contextual fitness planning.

EDUCATION

California State University, East Bay 2024 – Present
GPA: 3.9/4.0

B.Tech in Computer Science 2019 – 2023
Jawaharlal Nehru Technological University Hyderabad, India